Consumer Credit Risk Modelling

Loan Default Prediction with Canadian Macroeconomic Stress Testing for Credit Union Decision Support

Final Project Report

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Springboard Data Science Career Track — Capstone Three

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GitHub repository: github.com/MIJUMBO/consumer-credit-risk-modelling

Contents

Note: in Microsoft Word, right-click the table and select "Update Field" to populate page numbers.

Executive Summary

Canadian credit unions must hold forward-looking provisions against expected credit losses under IFRS 9, yet many smaller lenders still rely on flat, portfolio-average reserve rates that treat every borrower as carrying identical risk. This project builds an end-to-end, decision-ready credit risk pipeline that answers three questions a credit committee actually asks: **which borrowers are likely to default, how much should we provision today, and what happens to that provision if the economy deteriorates?**

Using 1.35 million resolved LendingClub consumer loans (2007–2018) as the borrower-level dataset and a five-indicator Canadian macroeconomic panel (Bank of Canada and Statistics Canada, 2000–2026) as the scenario layer, the project trains and compares three classifiers — Logistic Regression, XGBoost, and LightGBM — on a strictly leakage-free, origination-time feature set. A tuned XGBoost model is selected as champion (held-out test AUC **0.6726** on out-of-time 2017–2018 vintages), recalibrated with isotonic regression so its probabilities are valid dollar-weighted PDs (Brier improves from 0.2254 to **0.1576**, beating the 0.1676 base-rate benchmark), and explained with SHAP to confirm it relies on economically meaningful drivers — FICO score, loan-to-income, and debt-to-income.

The calibrated PDs feed an IFRS 9 Expected Credit Loss framework (ECL = PD × LGD × EAD) with three-stage classification. On the $242.1M test portfolio, the baseline allowance is **$56.8M (23.5% coverage)**. Stressing the portfolio with unemployment shocks anchored in Canadian history — a +2.7pp GFC replay and a +4.5pp beyond-sample severe scenario — raises required provisions by **+15.5% ($8.8M)** and **+29.0% ($16.5M)** respectively, driven primarily by performing loans migrating across the SICR threshold into lifetime-ECL Stage 2. The exercise demonstrates that the value of a PD model is not its AUC; it is the ability to allocate provisions according to risk, identify significant increases in credit risk, and quantify the capital buffer a downturn would demand.

1. Problem Identification

1.1 Context

Since 2018, IFRS 9 has required lenders — including Canadian credit unions — to recognise **expected** credit losses rather than incurred losses. Provisions must be forward-looking, must scale with credit deterioration through a three-stage classification, and must incorporate reasonable and supportable macroeconomic scenarios. For institutions without internal PD models, the practical fallback is a flat provision rate applied uniformly across the book. A flat rate may match the portfolio total in calm conditions, but it over-provisions low-risk borrowers, under-provisions high-risk ones, treats credit-impaired exposures no differently from performing loans, and offers no mechanism for translating an economic scenario into a provision number.

1.2 Problem Statement

Can a leakage-free machine learning model, trained only on information available at loan origination, produce calibrated probabilities of default that are accurate enough to (a) support IFRS 9 staging and provisioning, and (b) quantify how required provisions change under adverse Canadian macroeconomic scenarios — providing a credit union risk committee with a defensible, explainable alternative to flat-rate reserving?

1.3 Criteria for Success

1. A probability-of-default model that discriminates meaningfully better than random (AUC materially above 0.50) on a strictly **out-of-time** test set, with stable performance across loan vintages.
2. Calibrated probabilities suitable for dollar-weighted use: post-calibration Brier score better than a naive base-rate prediction, with mean predicted PD close to the observed default rate.
3. A complete IFRS 9 ECL implementation (staging, 12-month vs lifetime horizons, PD × LGD × EAD) producing a baseline provision and a stress-scenario provisioning table, with every scenario computed through one shared code path so changes are attributable solely to the applied shock.
4. Full model explainability (global and loan-level) sufficient for model governance and adverse-action reasoning.

1.4 Scope and Constraints

The project follows a deliberate three-layer architecture. **Layer 1** models borrower-level default risk on LendingClub data. **Layer 2** characterises Canadian macroeconomic conditions. **Layer 3** integrates the two at the portfolio level through stress scenarios and ECL. A critical architectural rule is enforced throughout: **the macro panel is never joined to loan-level data.** US borrower records carry no economically meaningful row-level link to Canadian indicators; the macro layer informs scenario design only. Key constraints: LendingClub provides resolved outcomes only through 2018; lifetime PDs must be approximated (no full term-structure data); and LGD, the lifetime multiplier, and the PD-unemployment elasticity are stated assumptions a lender would calibrate to internal loss history. The structure, not the parameter values, is the deliverable.

1.5 Stakeholders

The intended client is the credit risk function of a Saskatchewan credit union or community lender (e.g., institutions comparable to Conexus, Affinity, or Innovation Federal Credit Union): the Chief Risk Officer and credit committee (provisioning decisions), finance (IFRS 9 reporting), and model governance (validation and explainability requirements).

1.6 Data Sources

- **LendingClub accepted loans, 2007–2018Q4** — 2,260,701 loans × 151 columns, via Kaggle (`wordsforthewise/lending-club`). Borrower application, credit profile, and final loan outcome.
- **Bank of Canada Valet API** — overnight policy rate (series V39079 spliced with legacy V122530).
- **Statistics Canada** — national unemployment rate (table 14-10-0287), CPI all-items (18-10-0004), household debt-service ratio (11-10-0065), and consumer insolvencies (vector v41690973), all standardised to a monthly panel, 2000–2026 (318 months × 6 columns).

2. Data Wrangling

2.1 Target Construction and Outcome Filtering

Of 2.26M raw loans, only those with fully resolved outcomes were retained: Fully Paid, Charged Off, and Default. Active, late, in-grace, and policy-exception loans were excluded because their labels are unknown or biased. This produced **1,345,350 resolved loans** with a binary target `default_flag` (1 = Charged Off/Default) and an overall default rate of **19.96%** — a moderate 1:4 class imbalance handled later through class weighting rather than resampling.

2.2 Leakage Control: Two Distinct Exclusion Mechanisms

Two separate guardrails protect the model from learning information unavailable at origination:

- **LEAKAGE_COLUMNS (dropped entirely, 22 columns).** Post-outcome fields — repayment totals, recoveries, last payment dates, outstanding balances, post-origination FICO pulls — that only exist because the loan has already resolved. Training on these would inflate performance to meaningless levels.
- **MODEL_EXCLUDED_COLUMNS (retained for EDA, never modelled).** LendingClub's own risk assessments — `grade`, `sub_grade`, `int_rate` — are origination-time information, but including them would make the model predict the lender's pre-computed risk judgement rather than borrower risk: a circular prediction. They are kept in the dataset as EDA benchmarks only.

2.3 Cross-Vintage Feature Availability Audit

LendingClub's schema evolved between 2007 and 2018, so a feature present in 2017 may be entirely absent from 2010 training data. Every column's missingness was computed by vintage year and classified as CONSISTENT (<10% missing in all years), PARTIAL, LATE_ADDED (>50% missing pre-2013), or SPARSE (>40% missing overall). The audit classified **89 columns as SPARSE, 39 as CONSISTENT, and 4 as PARTIAL**. After removing leakage, target, and model-excluded fields and adding two controlled vintage features (`vintage_year`, `vintage_quarter` extracted from the issue date), **32 temporally stable feature candidates** survived. The full governance decision — consistent features, exclusions, leakage list, and analyst overrides — is exported to `consistent_features.json` for reproducibility.

2.4 Two-Stage Stratified Sampling

To balance iteration speed against benchmark reliability, two samples were drawn with stratification on `issue_year × default_flag` (preserving both the vintage distribution and the 19.97% default rate): a **100,000-loan development sample** for EDA, feature engineering, and model tuning, and a **500,000-loan production sample** reserved for the final champion retrain and out-of-time evaluation.

2.5 Canadian Macroeconomic Panel (Layer 2)

Five series were acquired programmatically — with chunked, memory-efficient ingestion and retry logic for the large Statistics Canada ZIP files — and standardised to monthly frequency (the quarterly debt-service ratio forward-filled). After boundary back/forward-filling of unpublished months, the unified panel spans **January 2000 to June 2026 with zero nulls** and is saved as `macro_panel.parquet`. It is used exclusively for scenario design, never joined to loans.

3. Exploratory Data Analysis

3.1 Borrower-Level Findings

- **Lender grade is strongly monotonic with default** (~6% for Grade A to ~49% for Grade G) — confirming both that the outcome is predictable and that grade must be excluded to avoid circularity.
- **Loan purpose separates risk meaningfully:** small business (27.7%), renewable energy (25.3%), and moving (24.2%) default most; car (16.0%), wedding (16.2%), and home improvement (17.3%) default least. This motivated an engineered three-tier purpose risk feature.
- **Vintage matters at the year level, not the quarter level.** Default rates climb from ~13% (2008–09, crisis-tightened underwriting and tiny volumes) to ~23% (2016–17, rapid platform growth and loosening standards). Calendar quarter spreads only 1.3pp — year-level vintage features carry the signal.
- **Distributional separation is real but modest:** defaulted loans show higher DTI and interest rates and lower FICO scores, with substantial overlap.
- **All individual correlations with default are weak** — FICO is the strongest protective feature (−0.129) and DTI the strongest risk feature (+0.093). Default is a non-linear, multi-factor outcome, which is precisely the case for tree-based models over a linear baseline.

3.2 Macroeconomic Findings

The macro panel tells one connected story: inflation and unemployment drive the Bank of Canada's policy rate, which in turn shapes household debt burdens. Two findings anchor the stress design. First, the largest sustained unemployment increase in the 26-year panel is the **2008–09 GFC: 6.1% → 8.8% trough-to-peak (+2.7pp)**. The COVID spike (+8.7pp) was larger but artificial and rapidly reversed — a poor template for sustained credit stress. Second, consumer insolvencies co-move with unemployment in normal times and through the GFC, but **decoupled during COVID** when CERB transfers and payment deferrals severed the usual link — the reason the PD-unemployment elasticity is treated as a scenario assumption, not an estimated law.

4. Feature Engineering and Pre-processing

4.1 The Fifteen-Feature Origination-Time Set

A single `engineer()` function — one source of truth applied identically to the development and production samples — produces the final feature matrix. The fifteen predictors group into five families:

- **Affordability:** loan_to_income, dti, loan_amnt, log_annual_inc (income log-transformed and 99th-percentile capped to tame skew).
- **Credit quality:** fico_orig (midpoint of the FICO range), revol_util (median-imputed), open_acc_band (ordinal: Few/Some/Many/VeryMany).
- **Derogatory flags:** delinq_flag, pub_rec_flag.
- **Employment:** employment_years (parsed from text), short_tenure_flag (<1 year).
- **Loan and vintage context:** purpose_risk_tier (Low/Med/High from observed EDA default rates), home_ownership, vintage_year, vintage_quarter.

A deliberate design lesson from the trial phase: DTI, FICO, and revolving utilisation enter as **raw continuous values** rather than the coarse bands used in earlier iterations — binning discarded signal and measurably lowered performance. Tree models choose their own split points.

4.2 Encoding and Standardisation

Three categoricals (open_acc_band, purpose_risk_tier, home_ownership) are encoded with an `OrdinalEncoder` fitted on the training split only and applied — never refitted — to validation, test, and the production sample, with unknown categories mapped to `-1`. Features were standardised with `StandardScaler` for Logistic Regression only (its optimisation is scale-sensitive); tree models split on thresholds and received raw values.

4.3 Temporal Train/Validation/Test Split

Loans are partitioned by origination year rather than randomly, so models are trained on history and evaluated on the future — exactly how a deployed model is used, and the only split that prevents vintage-level leakage from inflating metrics.

Split	Vintages	Rows (100k dev)	Default rate
Train	≤ 2015	61,442	18.43%
Validation	2016	21,786	23.29%
Test (out-of-time)	2017–2018	16,772	21.29%

Table 1. Vintage-based temporal split. The harder 2016 validation vintage (higher default rate) makes model selection a stress test of generalisation in itself.

5. Modelling

5.1 Three Models, One Evaluation Framework

Class imbalance (~1 default per 4.4 non-defaults) was addressed with `class_weight='balanced'` for Logistic Regression and `scale_pos_weight = 4.43` for the boosters. XGBoost and LightGBM were each tuned with a seeded, **time-boxed (30-minute) Optuna search** maximising validation AUC, with a 2,000-tree ceiling and early stopping determining the effective model size.

Model	Val AUC	KS	Gini	Brier
XGBoost (champion)	0.6684	0.2472	0.3367	0.2316
LightGBM	0.6684	0.2471	0.3369	0.2320
Logistic Regression	0.6616	0.2362	0.3232	0.2368

Table 2. Validation performance comparison. Base-rate Brier reference: 0.1786 at a 23.3% validation default rate.

5.2 Champion Selection: A Feature-Set Ceiling, Not a Tuning Limit

The two boosters tie to four decimals from entirely different configurations — XGBoost converged to **230 depth-4 trees**, LightGBM to **591 trees at 15 leaves** — and both optimisers actively preferred simpler, regularised models. Independent searches converging on the same score is evidence of a **stable predictive ceiling set by the leakage-free origination feature set**, not by algorithm choice. That ceiling is the intended cost of excluding the lender's own risk-pricing fields: a modest but honest level of discrimination. XGBoost is selected on parsimony (230 vs 591 trees — lighter to deploy and monitor) and a marginally better Brier score; Logistic Regression is retained as an interpretable benchmark.

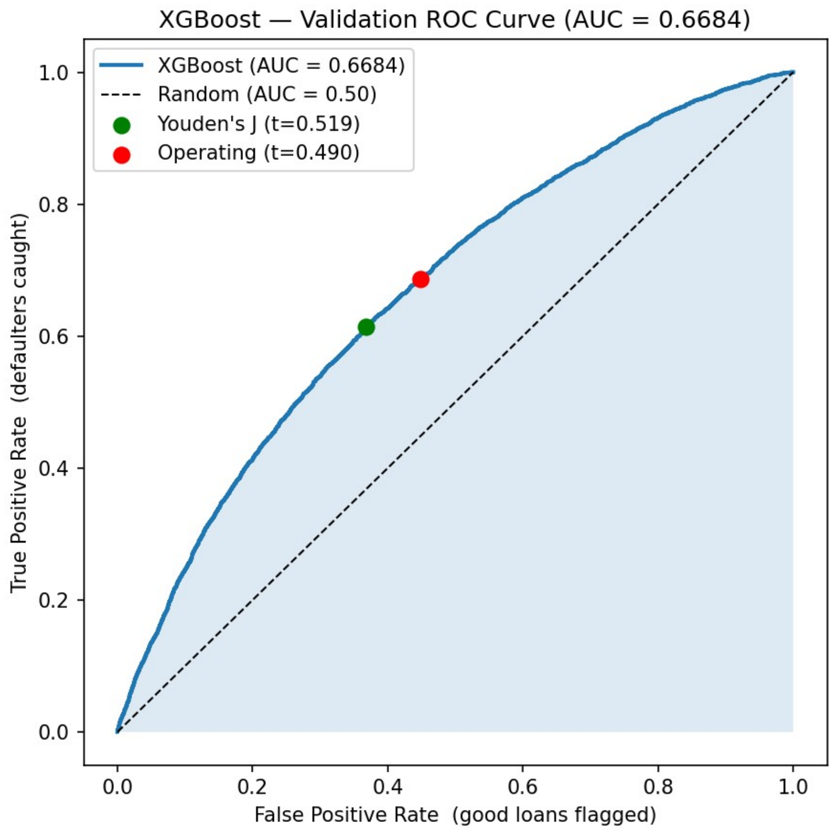

Figure 1. Champion validation ROC curve (AUC 0.6684) with the statistical optimum (Youden's J, t = 0.519) and the business operating point (t = 0.490) marked.

5.3 A Dollar-Weighted Operating Threshold

Rather than an arbitrary 0.50 cutoff, the diagnostic operating threshold was chosen by a risk policy: among all cutoffs capturing at least **75% of expected default dollars** (loss proxied as LGD × loan amount, so large defaults matter more), select the one declining the fewest creditworthy dollars. The resulting threshold (0.490) catches **3,483 of 5,073 future defaulters (69% recall)** at 32% precision and 58% accuracy — an intentional trade: in an imbalanced lending portfolio, accuracy is not the objective; missed defaults cost multiples of false alarms. This threshold is diagnostic only; the production IFRS 9 framework consumes continuous PDs.

5.4 Production Retrain and Temporal Drift

The champion configuration was retrained on the 500k production sample (307,208 training loans, vintages ≤ 2015) and evaluated once on the held-out 2017–2018 vintages (83,859 loans): **test AUC 0.6726**, slightly above validation. Scored vintage-by-vintage, discrimination is stable — 2017: 0.6725; 2018: 0.6668 — with the small 2018 dip within sampling noise on the smaller (~21k) cohort. The model generalises across loan cohorts rather than memorising one economic period.

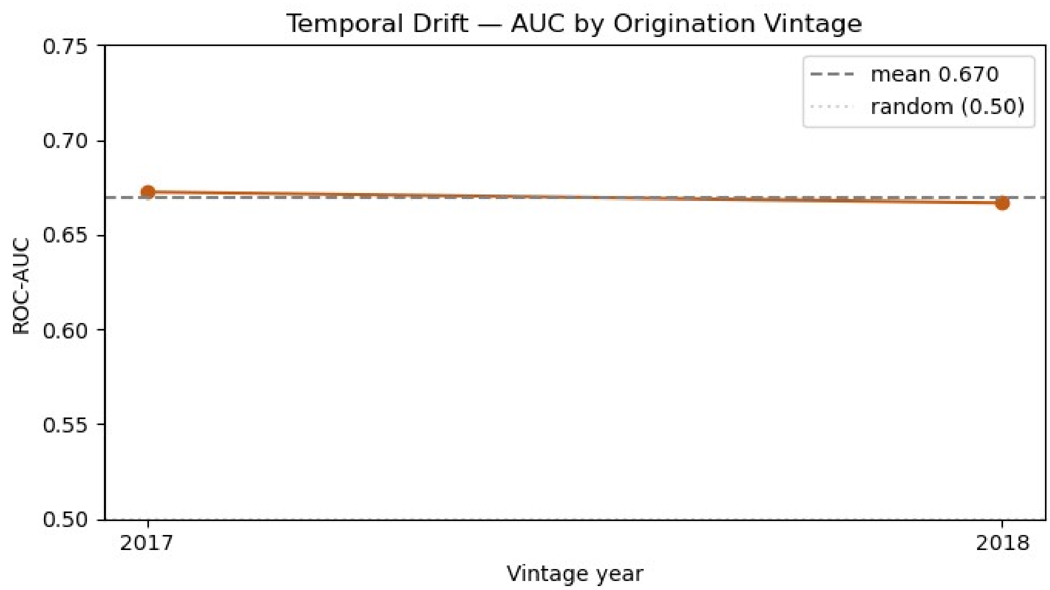

Figure 2. ROC-AUC by origination vintage on the held-out test window. Stable discrimination (~0.67) across cohorts is the primary evidence of production-grade generalisation.

5.5 Calibration: Turning Scores into Valid PDs

Class weighting inflates raw probabilities: the champion's raw Brier (0.2254) was **worse than a constant base-rate prediction (0.1676)** — loans scored around 55% PD actually defaulted ~26% of the time. Used directly in ECL, these PDs would overstate provisions materially. An **isotonic calibrator** was therefore fitted on the validation vintage (held out from production training). Because isotonic regression is monotonic, AUC, KS, and Gini are untouched; only the probability scale is corrected. Post-calibration Brier: **0.1576, beating the base-rate benchmark** — the single result that licenses multiplying these probabilities by dollars.

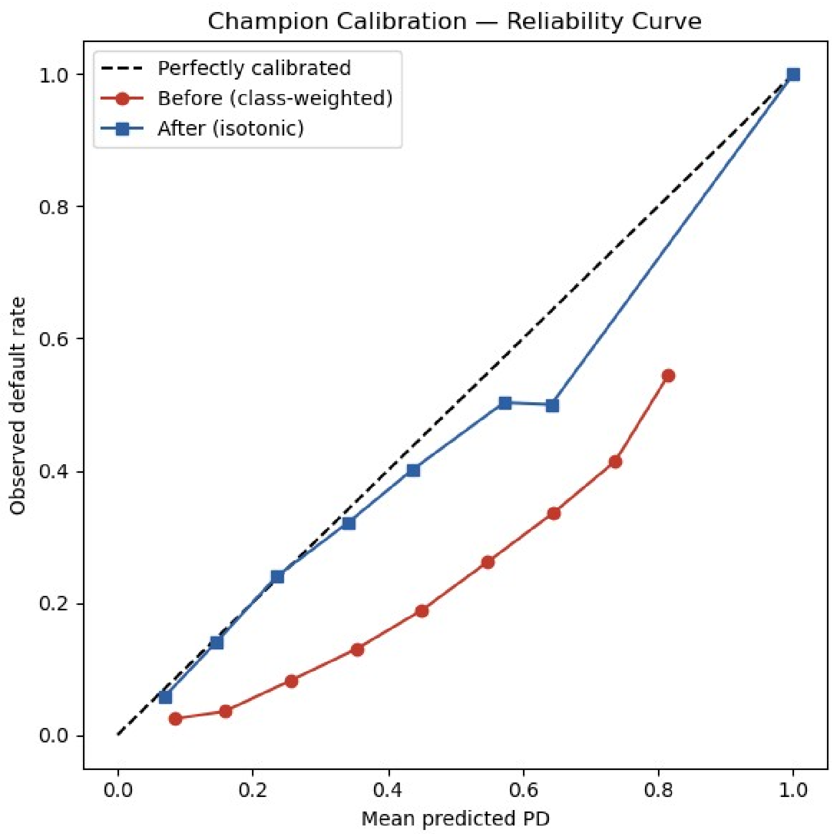

Figure 3. Reliability curve before (red) and after (blue) isotonic calibration. The calibrated curve hugs the 45° diagonal, making the PDs valid inputs to ECL = PD × LGD × EAD.

6. Model Explainability (SHAP)

A credit model that cannot be explained cannot be deployed: regulators and credit officers need to know why a borrower is flagged. SHAP values — additive and model-faithful — were computed with TreeExplainer on 5,000 validation loans. Where the gain-based importances of Section 5 explain how the model was **built**, SHAP explains how it **predicts** on unseen data; both rank the same core drivers, indicating a stable model.

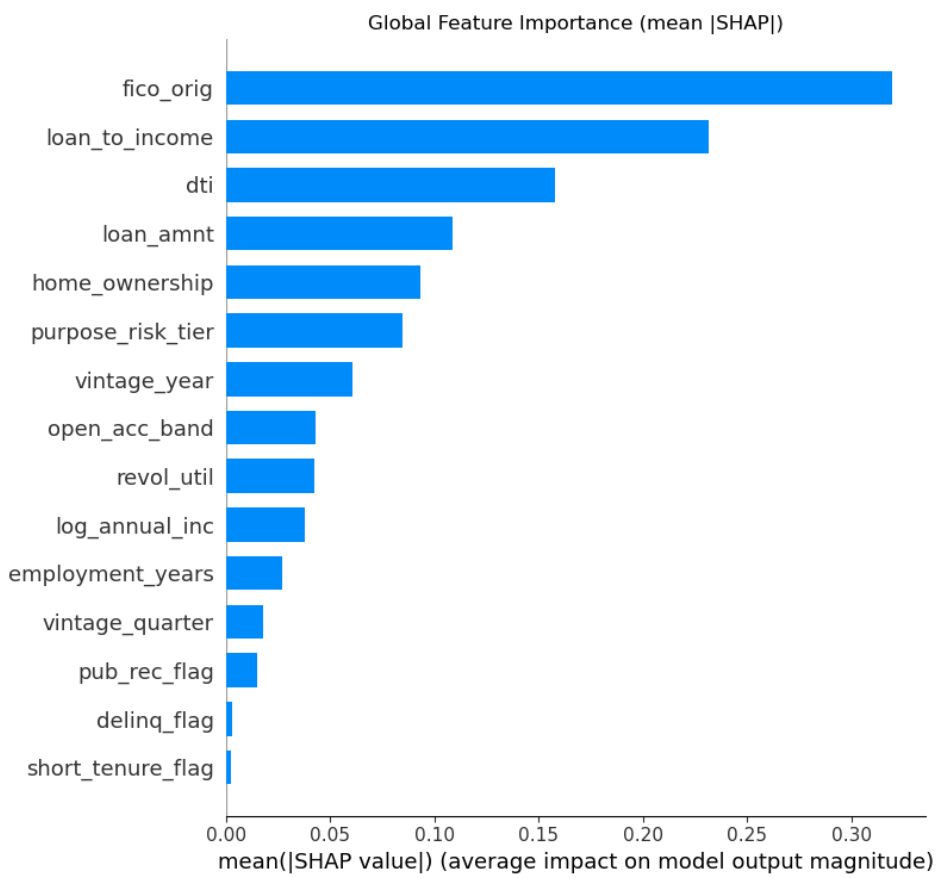

Figure 4. Global SHAP feature importance. Credit quality and affordability dominate: fico_orig (0.320), loan_to_income (0.231), dti (0.158), loan_amnt (0.109).

Three layers of evidence align. **Globally**, the model is driven by credit quality and affordability, with engineered features (home ownership, purpose risk tier, vintage year) contributing meaningfully. **Directionally** (beeswarm), higher leverage and indebtedness push risk up while stronger FICO, income, and tenure pull it down — consistent with credit theory. The FICO **dependence plot** shows a smooth, nearly monotonic decline in contribution as scores rise: a learned pattern, not isolated thresholds. **Locally**, a waterfall plot decomposes any single decision into exact feature contributions — the transparent, auditable rationale an adverse-action notice requires. Portfolio-level drivers reappear consistently in individual decisions, the property model governance looks for.

7. Stress Testing and IFRS 9 Expected Credit Loss

7.1 Scoring and Validation Gates

The calibrated champion scores the 16,772-loan held-out portfolio. Before any provisioning, two gates run loudly: the loaded artifact reproduces the production test AUC (right model, not a stale file), and the **mean calibrated PD (0.2205) sits beside the observed default rate (0.2129)** — the comparison that justifies dollar-weighting. Structural asserts (row counts, positive exposures, PDs in [0,1]) enforce the conditions every later cell silently relies on.

7.2 IFRS 9 Framework and Assumptions

ECL = PD × LGD × EAD, with **LGD = 45%** (typical unsecured consumer lending) and EAD proxied by loan amount. Staging: **Stage 1** (performing; 12-month ECL); **Stage 2** (significant increase in credit risk, proxied by calibrated PD above the 80th-percentile baseline threshold of 0.3064; lifetime ECL); **Stage 3** (credit-impaired — identified here by realised default since the test vintages are fully resolved; PD = 100%, so ECL = LGD × EAD). Lifetime PD is approximated as **2.5× the 12-month PD**, capped at 100% — a judgement validated in-notebook against the portfolio's term mix (78.5% 36-month, 21.5% 60-month loans imply a blended multiplier range of 2.32–2.59; 2.5 sits inside it, toward the conservative end). Critically, staging and ECL live in **one function, stage_and_ecl()**, reused unchanged across baseline and all stress scenarios — and a regression assert confirms the zero-shock scenario reproduces the baseline to the cent, so any provisioning change is attributable solely to the applied shock.

7.3 Baseline Allowance and the Value of the Model in Dollars

Stage	Loans	Exposure	ECL	Coverage
Stage 1 — Performing	10,687	$135,450,375	$10,828,689	7.99%
Stage 2 — SICR	2,514	$50,454,125	$20,708,922	41.05%
Stage 3 — Credit-impaired	3,571	$56,212,200	$25,295,490	45.00%
Total	16,772	$242,116,700	$56,833,101	23.47%

Table 3. Baseline ECL by IFRS 9 stage. Coverage rises sharply with credit deterioration — the graduated treatment a flat rate cannot produce.

A naive flat-rate reserve (one portfolio-average PD on every loan) would hold **$23.2M** against this book — **$33.6M (−59%) less** than the risk-sensitive allowance, because it treats credit-impaired exposures like performing loans and ignores the concentration of lifetime risk in the SICR segment. The absolute coverage level (23.5%) reads high by industry standards and is interpreted cautiously: the fully resolved test vintages contain 3,571 realised defaults provisioned at full LGD × EAD, so the **change in coverage across scenarios is more informative than the level**. The model's value is risk differentiation: directing provisions toward the loans most likely to generate losses while avoiding unnecessary reserving on low-risk exposures.

7.4 Macro Stress Overlay: Scenarios Anchored in Canadian History

Shock magnitudes come directly from the Layer 2 panel rather than round numbers. The **Adverse scenario (+2.7pp unemployment) replays the 2008–09 GFC exactly** — the largest sustained increase in the 26-year panel. The **Severely Adverse scenario (+4.5pp) is deliberately beyond-sample at ~1.7× the worst observed episode**, consistent with regulatory severe-scenario design. A stated elasticity links the shock to risk: each +1pp of unemployment raises every PD by 10% relative. Because the SICR threshold stays fixed at its baseline value, stress pushes loans across it — ECL rises through two channels at once: every PD lifts, and migrating loans switch from 12-month to lifetime ECL. That second, migration-driven channel is the non-linear behaviour IFRS 9's point-in-time staging is designed to capture.

Scenario	Shock	Total ECL	Coverage	Stage 2 loans	ECL vs baseline
Baseline	+0.0pp	$56,833,101	23.47%	2,514	+0.0%
Adverse (GFC replay)	+2.7pp	$65,636,032	27.11%	3,537	+15.5%
Severely Adverse	+4.5pp	$73,333,368	30.29%	5,022	+29.0%

Table 4. Provisioning under stress. Stage 3 stays fixed at 3,571 (impairment is realised, not scenario-dependent); the Stage 2 population roughly doubles under severe stress.

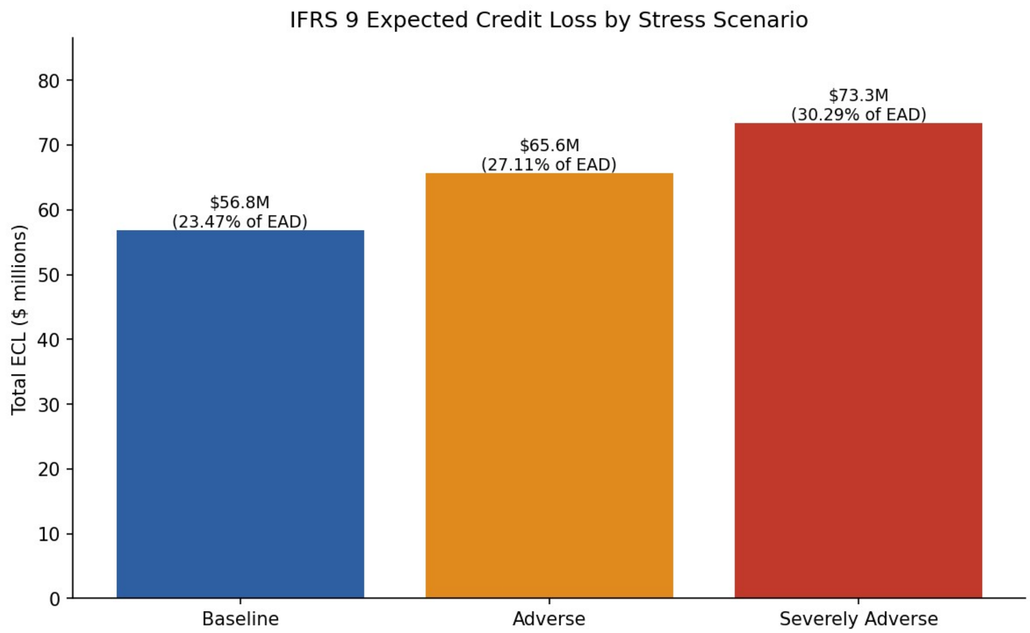

Figure 5. IFRS 9 Expected Credit Loss by stress scenario — the provisioning chart a risk committee reviews.

The response is broadly proportional to shock size because the framework applies a constant elasticity; real downturns often produce accelerating losses, so the severe-scenario figure should be read as a conservative illustration rather than a fully calibrated recession estimate.

8. Summary of Findings

1. **Discrimination is real, stable, and honestly bounded.** The champion achieves out-of-time test AUC 0.6726 with no temporal degradation across vintages. Two independent hyperparameter searches converging on identical validation AUC from different architectures shows the ~0.67 ceiling belongs to the leakage-free origination feature set — the deliberate price of excluding the lender's own risk pricing.

2. **Calibration, not AUC, is what makes a PD model bankable.** Raw class-weighted probabilities were worse than a constant base-rate guess (Brier 0.2254 vs 0.1676) and would have overstated provisions; isotonic recalibration (Brier 0.1576) made them valid dollar weights without touching rank order. The model is not the product — the provision number is.

3. **The model's drivers are economically defensible.** SHAP confirms FICO, loan-to-income, and DTI dominate, with directions consistent with credit theory, traceable from portfolio level down to individual lending decisions.

4. **Risk-sensitive provisioning changes the answer materially.** Baseline allowance: $56.8M (23.5% coverage) vs $23.2M flat-rate. Under a GFC-replay shock provisions rise $8.8M (+15.5%); under a beyond-sample severe shock, $16.5M (+29.0%) — driven primarily by a doubling of the lifetime-ECL Stage 2 population (2,514 → 5,022 loans).

9. Recommendations to the Client

1. **Adopt model-based, staged provisioning in place of flat-rate reserving.** The calibrated PD pipeline supports IFRS 9 staging end-to-end and prices risk where it actually sits: 8.0% coverage on performing loans, 41.1% on SICR exposures, 45.0% on impaired loans. Begin with a parallel run alongside the current approach, reconciling the two allowances quarterly before cut-over.

2. **Pre-position a stress buffer and monitor SICR migration as the early-warning metric.** Board-level capital planning should contemplate an additional $8.8M of provisions in a GFC-severity downturn and $16.5M in a severe one (on a $242M book; ~3.6–6.8% of EAD). Because the provisioning response is driven by Stage 1 → Stage 2 migration, the count of loans approaching the SICR threshold is the metric to track monthly — it moves before losses do.

3. **Recalibrate the stated assumptions to internal data before production use.** LGD (45%), the lifetime multiplier (2.5×), and the PD-unemployment elasticity (10% per pp) are industry-standard placeholders. Replace them with figures estimated from the institution's own loss history and Canadian member data, and re-fit the isotonic calibrator on the live book — the framework is built so only these constants change.

10. Ideas for Further Research

- **Lift the feature-set ceiling with richer data.** Credit bureau trade-line attributes, behavioural/transactional data, or alternative data would attack the binding constraint directly — the 0.667 AUC plateau is a data problem, not a model problem.
- **Model the lifetime PD term structure properly.** Replace the 2.5× approximation with survival analysis (Cox or discrete-time hazard models) to estimate month-by-month default hazards, enabling exact lifetime ECL for Stage 2.
- **Estimate, rather than assume, the macro elasticity.** Regress vintage-level default rates on the Canadian panel (with the COVID decoupling explicitly handled) and allow non-linear or regime-dependent responses so severe scenarios capture loss acceleration.
- **Production hardening.** Population Stability Index monitoring, scheduled recalibration, champion–challenger governance, and fairness testing across protected-attribute proxies before any lending use.
- **Canadian-native validation.** Partner with a credit union to re-estimate the pipeline on domestic loan performance data, closing the geographic gap between the US training book and the Canadian deployment context.

Appendix: Repository and Reproducibility

All code, notebooks, figures, and saved model artifacts are available at github.com/MIJUMBO/consumer-credit-risk-modelling. The pipeline runs as six sequential notebooks:

- `01_data_preparation.ipynb` — acquisition, target construction, leakage control, vintage audit, sampling, macro panel build.
- `02_eda_loans_macros.ipynb` — borrower-level and macroeconomic EDA.
- `03_feature_engineering_preprocessing.ipynb` — engineer() function, 15-feature list, temporal split, encoding, dataset export.
- `04_modeling.ipynb` — three-model comparison, Optuna tuning, champion selection, threshold policy, production retrain, drift analysis, isotonic calibration.
- `05_model_explainability.ipynb` — SHAP global, directional, dependence, and waterfall analyses.
- `06_stress_testing_and_ecl.ipynb` — IFRS 9 staging, baseline ECL, naive benchmark, macro stress overlay, provisioning table.

Key saved artifacts: `champion_xgb_production.pkl` (raw model for SHAP/ranking), `champion_xgb_calibrated.pkl` (calibrated PDs for ECL), `ordinal_encoder.pkl`, `feature_list.txt`, `consistent_features.json`, and the processed parquet splits. All random operations are seeded (random_state = 42); the Optuna searches use seeded TPE samplers for reproducibility within the stated time budget.